CSE3063 JAVA PROJECT: Iterative analysis, design and implementation of a Course Registration and Tracking System

We are in need of a software for registration of courses and tracking of courses in a distance education setting. You need to create courses modelling after the courses in our department. There will be prerequisites of the courses and other business logic will also be modelled after our departments rules and regulations for taking courses. In order to simulate your software and test your business logic, there will be students in your software that randomly generated that can be in each of the 8 semesters with equal probability. When you are creating a student for a particular semester, you will choose random number of courses and she will be marked failed from these courses. You need to consider the prerequisite tree in implementing this starting from the 3rd semester. You need to store these students and their transcripts in json files. There will be a json file for each student. In order to simulate course registration, process you will either choose a randomly selected student or the json file name of a student. In your user interface (for your first iteration you will have a command line interface only) you will list the course that she can take based on the business logic (rules and regulations of the department and prerequisites). She will register the elective courses randomly from a pool of courses. You need to show each step on the user interface one by one. With a certain probability an elective course's quota may be full. In your simulation, in addition of progressing the registration steps one by one with the user input, you should also have an option to batch process for a single student to register for all the courses for all semester from the first semester to her graduation and output all the registration actions to the screen and a json log file with the student's name.

Rules and Requirements:

You will be graded using bell curve. Therefore, all groups are in competition.

There will be no databases in this project. You can't use a database system at all. You may store your data in json files.

You must use Java in an Object-Oriented Manner as we introduce in the lectures. However, I want you to go beyond what we teach, do your research on the internet, try to apply best practices and design patterns. You will be grading based on your object oriented modeling.

You must use iterative and evolutionary development process such as UP, SCRUM, XP. Therefore, you need to start with the core requirements in the first iteration and add more functionality at each subsequent iteration.
When preparing analysis and design artifacts such as UML diagrams you must follow the instructions exactly and spend more than usual time in analysis and design phases.
Your business logic must completely be separated from the user interface.
The main problem is we don't have all the requirements in the beginning. Rest of the requirements will be revealed during next iterations.
Therefore, you must put special emphasis on the extensibility of your code. This will be evaluated by looking at the change between two iterations. Change will be measured in terms of following metrics.

You need to measure and report these at each new iteration. lines of code (LOC) added (previous iterations LOC – this iterations LOC), number of changed classes, number of changed methods, number of newly added classes, number of newly added methods, number of newly added attributes A more detailed document explaining the evaluation criteria will be provided separately.

You must implement the data labeling system as a simulation for the first iteration.
You should read student information as a json file.
Your code must support easily pluggable labeling mechanisms. (Hint: A nice place to apply Polymorphism and related design patterns!)
If we integrate your business logic code to a user interface (think about the user interface labeling mechanism above) it should work with minimal changes. So, at the end of the project you will be developing a simple user interface to demonstrate this. You need to report above metrics to prove this.

After you determined your project groups, you need to

1- open a GitHub account with a meaningful, identifiable name e.g. murat.ganiz or mcganiz

2- open a private GitHub repository using the following naming pattern: CSE3063F21p1[group number]

For group 1, repo name should be: CSE3063F21P1_GRP1

(note: do not open a organization! just open a repo. One of the group members should open the repo and add others as collaborators)

3- add your fellow group members as collaborators to this repo

4- add me (murat.ganiz or mcganiz) as collaborator to this repo

5- create and empty project with a single java file which prints "hello world" as soon as possible and push it to the repo.

6- Add a readme file, fill it with your group number and list the numbers and names of your group members.

7- Open a folder for each iteration like “iteration1” and store your iteration code in the corresponding folder.

Push (Please see Figure 1 below) as frequently as possible and meaningful! All students in the group must push.

You need to provide following artifacts by the iteration deadlines:

1- A simple Requirement Analysis Document (RAD)

This may include a brief description of the project in a paragraph or two, glossary, list of functional and non-functional requirements, a domain model showing real world objects in your domain along with their simple relations and features. Optionally you may also add a System Sequence Diagram (SSD).

(File name example: CSE3063F21P1_RAD_GRP1 _iteration1.pdf)

This must be pushed into your GitHub repository’s corresponding iteration folder by the deadline.

2- Design Class Diagram (DCD)

An UML class diagram showing
your domain classes. C
(File name example: CSE3063F21P1_DCD_GRP1_iteration1.pdf)
This must be pushed into your GitHub repository's corresponding iteration folder by the deadline.

3- Design Sequence Diagrams (DSD)

UML Sequence Diagram(s)
showing interactions between software objects of your system. (File name example: CSE3063F21P1_DSD_GRP1_iteration1.pdf)
This must be pushed into your GitHub repository's corresponding iteration folder by the deadline.

4- Java Code

This includes classes. Each class must be in a separate .java file named with the name of the class.
Please make sure that only the
java files, json files and necessary libraries are in the repo. Your repo size should not exceed 10mb. During your demonstrations me or your TA should be able to run your java code easily.
This must be pushed into your
GitHub repository's corresponding iteration folder by the deadline.

5- Project Evaluation Metrics

This is a spreadsheet (in
excel or open office spreadsheet file) providing your measurements for the metrics given in "CSE3063 Fall2021 Project Evaluation Metrics" file. The metrics will be given as columns. Please see the example spreadsheet material under Java Project topic in classroom.
(File name example: CSE3063F21P1_Metrics_GRP1_iteration1.ods)

Please make a
compressed zip archive of all these 5 artifacts and upload it to Google classroom by the deadline. Best way to do this is to take a snapshot of your GitHub repo and zip it. File size should not exceed 10mb. All students in a group should upload the same file to google classroom.